

Quiet Sentinel

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FEATURES

Creating a Rails App Using JumpStart Pro

Scott Johnson (fuzzyblog)

The view of the shelling of Kharkiv at 4 am

Donate Now to Pizza for Ukraine

The following blog post is an excerpt from a series of articles I'm working on about JumpStart Pro. I found myself referring to the steps below that I thought it might be generally useful for people.

I'm going to walk you through the "post purchase" experience with Jump Start Pro (JSP). This will guide you through creating a new app using JSP.

Step 1: Cloning an App Into Your Github

Here's what you do:

Purchase your Jumpstart Pro license.

Log in to <https://Jumpstartrails.com/> if you haven't already.

Click on your license name.

You will end up at a page which tells you to visit your Github repo. You need to goto: <https://github.com/orgs/Jumpstart-pro> . This in turn will take you here: <https://github.com/Jumpstart-pro>

Click into the Rails repository <https://github.com/Jumpstart-pro/Jumpstart-pro-rails>

Click the green Use This Template button.

Change the owner to be your github name

Enter the name of the NEW repo:

Set your description if you care (I never do).

Change the setting to be private .

I checked off ☐ include all branches (no clue if that was right or not)

Click the green Create repository from template button

Step 2: Getting that Cloned App Onto Your Machine

Github is going to clone the Jumpstart pro rails repo over to your personal github account. Here's what's next

On your new repo, click the green Code drop down and get the clone url .

On your development machine, change to your local development directory and clone the repo: `git clone` (what you copied)

Set up your local ruby to be 3.1 if you don't have it.

Run a bundle install

Edit `config/database.yml` and change database names as needed.

Now you want to run:

`bin/rails db:create`

Step 3: Setup and Getting to Green Tests in a Rails 7 World

When you're starting with something complex that someone else built then tests are always, always, always your friend. So this means that you start with:

`bin/rails test`

This is going to fail but we're working from failure to success (a red / green refactor approach).

Run `npm install`

Run `yarn install --check-files` (Disclaimer – the whole asset pipeline largely remains a mystery to me; I think this is needed but, for all I really know, running it could kill Santa's elves and invoke the apocalypse).

Run `bin/rails test` - if this passes, GREAT! But if you get the error: `DRb::DRbRemoteError: The asset "application.js" is not present in the asset pipeline.` then revert your node version back to 16.

Here's how to check your node version:

`> node -v v18.2.0`

Just like ruby has RVM, node has NVM (node version manager) so install that.

`nvm install 16.3.0`

and then:

`npm install`

Now you need to build the JavaScript for the app:

`rails javascript:build`

The next error you will get, if you try to test again, is:

The asset "application.css" is not present in the asset pipeline

so now you need to do:

`rails css:build`

And now you can do:

`bin/rails test`

Filtering as domain logic

Mark Seemann · Mark Seemann (ploeh blog)

Performance and correctness are two independent concerns with overlapping solutions.

How do you design, implement, maintain, and test complex filter logic as part of out-of-process (e.g. database) queries?

One option is to implement parts of the filtering logic twice: Once as an easily-testable in-memory implementation to ensure correctness, and another, possibly simpler, query using the query language (usually, SQL) of the data source.

Does this not imply duplication of effort? Yes, to a degree it does. Should you always do this? No, only when warranted. As usual, I present this idea as an option you may consider; a tool for your software design tool belt. You decide if it's useful in your particular context.

Motivation

When extracting data from a data source, an application usually needs some of the data, but not all of it. If the software system in question has a certain size, the subset required for an operation is only a miniscule fraction of the entire database. For example, a user may want to see his or her latest order in a web shop, but the entire system contains millions of orders. Another example could be a system for managing help desk requests: Each supporter may need a dashboard of open cases assigned to him or her, but the system holds millions of tickets, and most of them are closed.

If a data store supports server-side querying, for example with SQL or Cypher , it's reasonable to let the data store itself do the filtering.

As anyone who has worked professionally with SQL can attest, SQL queries can become complicated. When this happens, you may become concerned with the correctness of a query. Does it include all the data it should? Does it exclude irrelevant data? If you later change a query, how can you verify that it still works as intended? How do you even version it?

Automated testing can address several of these concerns, but testing against a real database, while possible, tends to be cumbersome and slow. Do alternatives, or augmentations, exist?

How it works

If a server-side query threatens to become too complicated, consider shifting some of the work to clients. You may retain some filtering logic in the server-side query, but only enough to keep performance good, and simple enough that you are no longer concerned about its correctness.

Implement the difficult filtering logic in a client-side library. Since you implement this part in a programming

language of your choice, you can use any tool or technique available in that context to ensure correctness : Test-driven development, static code analysis, type checking, property-based testing, code coverage , mutation testing, etc.

Using a funnel as a symbol of filtering, this diagram depicts the idea:

Normally a funnel is only useful when the widest part faces up, but on the other hand, we usually depict application architectures with the database under the application. You have to imagine data being 'sucked up' through the funnels.

In reality, the two filters will differ, but have overlapping functionality.

If based on a relational database, the server-side query will still hold table joins and column projections that are effectively irrelevant to the client-side Domain Model. On the other hand, while the server-side query may apply a rough filter, the more detailed selection of what is, and is not, included happens in the client.

The server-side query is defined using the query language of the data store, such as SQL or Cypher. The client-side query is part of the application code base, and written in the same programming language.

When to use it

Use this pattern if a server-side query becomes so complicated that you are concerned about its correctness, or if correctness is an essential part of a Domain Model's contract.

While it is conceptually possible to load the entire data store's data into memory , this is often prohibitively expensive in terms of time and memory. It is often necessary to retain some filtering logic (e.g. one or more SQL WHERE clauses) on the server to pare down data to acceptable sizes. This implies a degree of duplicated logic, since the client-side filter shouldn't assume that any filtering has been applied.

Duplication comes with its own set of problems, even if this looks like the benign kind . Alternatives include keeping all logic on the database server, which is viable if the logic is simple, or can be sufficiently simplified. Another alternative is to perform all filtering in the client, which may be an attractive solution if the entire data set is small.

Encapsulation

If a Domain Model is composed of pure functions , data must be supplied as normal input arguments. In a more object-oriented style, data may arrive as indirect input . In both object-oriented and functional architecture, encapsulation is important . This entails being explicit about invariants and pre- and postconditions; i.e. contracts .

To enforce preconditions a Domain Model must ensure

Deploying to HatchBox with Rails 5.x

Scott Johnson (fuzzyblog)

I just got a database less home page for a future SAAS app deployed via HatchBox under Rails 5.x and I thought the differences were significant enough that I should document it. HatchBox is the best tool I've ever found for Rails deploys but knowing all the steps to follow is key.

elixi: Other than the Rails 5.2 upgrade step, there aren't many differences between this and Rails 6 but it has been long enough since I've done this that writing it down felt right.

1. Upgrade Rails to "5.2.x" if You Don't Have rails credentials:edit

If your Rails 5 version is old enough that you get this crap:

➤ rails credentials:edit rails aborted! Don't know how to build task 'credentials:edit' (see --tasks)

Here's the line you want to put into your Gemfile:

Once you change your Rails version, you want to use:

bundle update

to upgrade your Rails installation.

2. There is No True Database Less Deploy in HatchBox

HatchBox is designed to work with a database (which is also the Rails default) because it calls database create and database migrate as part of the deploy. The trick here is to use an existing database and set your env variables and database.yml to allow for this

Here's how to set up your database.yml file:

```
production: adapter: postgresql url: encoding: utf8 pool: 50
```

Your env variables are described in the section below.

As long as you don't have migrations or database tables as part of your app, you can use any DATABASE_URL value (you set this below under environment variables; I just copied one from a different application).

4. Run rails credentials:edit

The command:

```
rails credentials:edit
```

creates the Rails master key and secret key base that you need.

You want to save your master key and secret key base because you need both in the env variables section.

5. HatchBox Setup

You want to:

Add your app to github or another supported version control system and note the username/repo_name pairing

Create a new app

Choose the cluster

Add your env variables (see next section)

6. Set Your Env Variables Correctly

Here are the env variables you want to set:

DATABASE_URL

RACK_ENV

RAILS_ENV

RAILS_MASTER_KEY

REDIS_URL

SECRET_KEY_BASE

7. Do a Deploy

After you get everything setup you want to do a deploy. This is necessary before you can do the SSL stuff.

8. Setup Your Domain

I added my domain to the Domains section of HatchBox both as the base domain and as the www variant.

9. Setup SSL

The next step is to select the SSL option and then turn on Let's Encrypt.

10. Deploy and Verify

There are two final steps. At this point everything should be setup so you want to do a final deploy and then verify your site in https mode.

11. Turn on Push Based Deploying

The absolute last step is to turn on git push based deploys so any code changes you make automatically trigger a deploy of your code. HatchBox makes this brilliantly simple:

Go to your app's main page.

Click the Deploy menu item.

Click the Green Automatic Deploy button

And then make a simple code change and push the code. At that point a deploy will start automatically.

yaml to allow for this Here's how to set up your database.

Scott Johnson (fuzzyblog)

What is Machine Learning?

Scott Johnson (*fuzzyblog*)

I have written a number of posts about different aspects of Machine Learning but, it occurs to me, not a single one that actually says “Machine Learning is xyz and here are the definitions” so here goes.

Machine Learning is a technology that allows a computer (“the Machine”) to take a “corpus” of “labeled” “training data” and then “learn” how to do something that the training data teaches the computer.

So the above is my definition but let’s look at Wikipedia:

Machine learning (ML) is the scientific study of algorithms and statistical models that computer systems use to perform a specific task without using explicit instructions, relying on patterns and inference instead. It is seen as a subset of artificial intelligence. Machine learning algorithms build a mathematical model based on sample data, known as “training data”, in order to make predictions or decisions without being explicitly programmed to perform the task.[1][2]:2 Machine learning algorithms are used in a wide variety of applications, such as email filtering and computer vision, where it is difficult or infeasible to develop a conventional algorithm for effectively performing the task. Wikipedia

And, while it is a simplification, the underlying thing idea driving machine learning is that of data driven pattern recognition .

Basically the Machine Learning process amounts to:

Take a bunch of data and construct a set of examples that illustrate what you want the machine to learn. This is called Training Data and the overall collection amounts to a Corpus .

Add metadata to these examples in a way that represent what you want the machine to learn. Examples could be blocks of text, images, video files or whatever. I personally work mostly in text but I have machine learning buddies who do things learning on images of cancerous masses. This is called Labeling . Labeling might consist of different things like adding Yes / No / Abstain values to a tweet or even extracting information from a recipe like the ingredients.

Running a computer program, the training program , against the data to generate a “neural network”, a computational engine that works much like the associative memory in our brains. And, just as we don’t understand how our brains work, we often don’t understand how a neural network works. This is known as the explainability problem in Machine Learning and I talk about it here .

Now, you have to be careful, because this is a simplification of a very complex technology but the above is the essence of Machine Learning.

Each of the key terms above, as well as a few others relevant to Machine Learning is described below.

GPU

A GPU or “Graphics Processing Unit” is a very, very powerful computer chip that is needed to run the training program that trains the neural network. Without a GPU, creating the neural network can literally take weeks. Wikipedia .

A neural network is an approach to computation that makes a computer work more similarly to a human brain. Wikipedia

This is a fancy damn word that means “related collection of stuff”. Generally Machine Learning refers to a textual corpus but it could be a collection of images.

Labels are the values applied to training data that represent what you want the machine to learn.

Labeling is the process of adding labels to the training data.

The labeler is the software that makes applying labels to the training data easier. A labeler might be a special computer program or even Microsoft Excel or something like AirTable.

The training set is the collection of data that you use to generate your models. The amount of data in a training set varies but it can often be as much as tens of thousands of labeled examples.

The test set is the collection of labeled data that you use to make sure that your trained model learns the way you think it does. After you train your model using the training set then you benchmark your model against the test set. Depending on how you build your models there can actually be many different sets.

Precision / Recall / F1 / AUC and Other Weird Words

These are terms that are used to measure how your model performs and are different mathematical approaches for model performance.

No More Blue Fridays

Brendan Gregg

In the future, computers will not crash due to bad software updates, even those updates that involve kernel code. In the future, these updates will push eBPF code.

Friday July 19th provided an unprecedented example of the inherent dangers of kernel programming, and has been called the largest outage in the history of information technology. Windows computers around the world encountered blue-screens-of-death and boot loops, causing outages for hospitals, airlines, banks, grocery stores, media broadcasters, and more. This was caused by a config update by a security company for their widely used product that included a kernel driver on Windows systems. The update caused the kernel driver to try to read invalid memory, an error type that will crash the kernel.

For Linux systems, the company behind this outage was already in the process of adopting eBPF, which is immune to such crashes. Once Microsoft's eBPF support for Windows becomes production-ready, Windows security software can be ported to eBPF as well. These security agents will then be safe and unable to cause a Windows kernel crash.

eBPF (no longer an acronym) is a secure kernel execution environment, similar to the secure JavaScript runtime built into web browsers. If you're using Linux, you likely already have eBPF available on your systems whether you know it or not, as it was included in the kernel several years ago. eBPF programs cannot crash the entire system because they are safety-checked by a software verifier and are effectively run in a sandbox. If the verifier finds any unsafe code, the program is rejected and not executed. The verifier is rigorous – the Linux implementation has over 20,000 lines of code – with contributions from industry (e.g., Meta, Isovalent, Google) and academia (e.g., Rutgers University, University of Washington). The safety this provides is a key benefit of eBPF, along with heightened security and lower resource usage.

Some eBPF-based security startups (e.g., Oligo, Uptycs) have made their own statements about the recent outage, and the advantages of migrating to eBPF. Larger tech companies are also adopting eBPF for security. As an example, Cisco acquired the eBPF-startup Isovalent and has announced a new eBPF security product: Cisco Hyper-shield, a fabric for security enforcement and monitoring. Google and Meta already rely on eBPF to detect and stop bad actors in their fleet, thanks to eBPF's speed, deep visibility, and safety guarantees. Beyond security, eBPF is also used for networking and observability.

The worst thing an eBPF program can do is to merely consume more resources than is desirable, such as CPU cycles and memory. eBPF cannot prevent developers writing poor code – wasteful code – but it will prevent serious issues that cause a system to crash. That said, as a new technology eBPF has had some bugs in its management code, includ-

ing a Linux kernel panic discovered by the same security company in the news today. This doesn't mean that eBPF has solved nothing, substituting a vendor's bug for its own. Fixing these bugs in eBPF means fixing these bugs for all eBPF vendors, and more quickly improving the security of everyone.

There are other ways to reduce risks during software deployment that can be employed as well: canary testing, staged rollouts, and "resilience engineering" in general. What's important about the eBPF method is that it is a software solution that will be available in both Linux and Windows kernels by default, and has already been adopted for this use case.

If your company is paying for commercial software that includes kernel drivers or kernel modules, you can make eBPF a requirement. It's possible for Linux today, and Windows soon. While some vendors have already proactively adopted eBPF (thank you), others might need a little encouragement from their paying customers. Please help raise awareness, and together we can make such global outages a lesson of the past.

Authors: Brendan Gregg, Intel; Daniel Borkmann, Isovalent; Joe Stringer, Isovalent; KP Singh, Google.

Test Amazon Alexa Skills Offline With Mocha And Chai For Node.js

Nic Raboy · Nic Raboy (polyglot developer)

By now you're probably aware that I'm all about Amazon Alexa skills since I'm a proud owner of an Amazon Echo. I had released a Alexa skill called BART Control and published a guide on creating a simple skill with Node.js and Lambda . If you went through my Node.js and Lambda guide you probably found it pretty painful to test the skill you were working on. The constant building and uploading to Lambda could easily get out of control. What if I told you there was a much simpler way that could save you a ton of time?

We're going to take a look at adding test cases for testing an Alexa skill offline without ever having to upload the skill to Lambda .

The post Test Amazon Alexa Skills Offline With Mocha And Chai For Node.js appeared first on The Polyglot Developer .

Tour Of Heroes As A NativeScript With Angular Android And iOS Application

Nic Raboy · Nic Raboy (polyglot developer)

When it comes to learning Angular, the go-to example is the Tour of Heroes tutorial that is found in the official Angular documentation. This is a great tutorial because it covers a lot of topics, however, it is a web application tutorial. What if we wanted to build a mobile application from it, or more specifically, a native mobile application with Android and iOS support?

We're going to see how to build a Tour of Heroes inspired iOS and Android mobile application using Angular and NativeScript .

The post Tour Of Heroes As A NativeScript With Angular Android And iOS Application appeared first on The Polyglot Developer .

IN BRIEF

IPFS Blog, IPFS Blog: The IPFS Newsletter is back, with many exciting updates to share: HTTP support across the IPFS stack, P2P in browsers, Kubo speedup, and more.

Justin Carter, Gizmodo: Whether you're excited for 'Exodus' or 'The Expanse: Osiris Reborn,' it looks like RPG fans will be eating good next year.

Nic Raboy, Nic Raboy (polyglot developer): I currently have a few Ionic Framework Android and iOS applications on the market. A common request for these apps have been to have a pin code unlock feature for extra security.

The post Add Pin Code Unlock To Your Ionic Framework App appeared first on The Polyglot Developer .

Nic Raboy, Nic Raboy (polyglot developer): I am pleased to announce my first Unity3D plugin is now available for download. EZ Analytics for Android offers Unity3D developers a native solution for Google Analytics with as little as two lines of code.

The post EZ Analytics for Android Released appeared first on The Polyglot Developer .

Cerner Engineering, Cerner Engineering: Currently in the world of JavaScript these options are what we most commonly use to generate logs: console.log console.info console.warn console.error These are actually pretty good in most modern browsers. Even if you go back to Internet Explorer 8 console.log and friends work as long as you have the developer tools open. Given that we have these logging utilities what is the problem with using them? When in local development these are just fine for helping debug and speed up development.

Nic Raboy, Nic Raboy (polyglot developer): I've recently been exploring other options when it comes to web frameworks. I come from a PHP ZendFramework background, but since having upgrading to ZendFramework 2, I've not been impressed.

Since then I've been exploring more of Node.js. However, like PHP, web programming often works best with a framework. This is why I've been exploring Node.js with Express as the framework.

The post Create A Simple Web Application With Express.js appeared first on The Polyglot Developer .

Nic Raboy, Nic Raboy (polyglot developer): So you've been fiddling with React Native for a bit now like I have. If you're like me then you're at a point where you're ready to look at saving and loading data in your mobile application. I'm going to show you how to do this with the help of the AsyncStorage module. This is a simple module that allows you to store and retrieve data from your device's local storage. It's a great way to persist data between app launches and offline use. In this post, I'll show you how to use AsyncStorage to save and load data in your React Native application. You'll see how to install the module, create some data, and then save it to the device's local storage. Finally, you'll see how to retrieve the data and use it in your application. This is a simple but powerful way to persist data in your mobile app. I hope you find this post helpful. If you have any questions, feel free to leave a comment below. Thanks for reading!

Updating to the Latest Version of JumpStart Pro

Scott Johnson (fuzzyblog)

So I've been using JumpStart Pro to build a new side project and in a discussion with Chris Oliver, the creator of JumpStart Pro, I reported a bug which he said "Oh - you should update to the latest". He viewed the update process as something seamless and I, honestly, didn't get it. And, in researching this, I discovered that I made a fatal mistake. The rest of this blog post covers that mistake and explains how you update JumpStart Pro. Here's the tldr (too long; did not read):

ALWAYS, ALWAYS, ALWAYS create a new app based on JumpStart Pro by cloning it from the GitLab source i.e. you start with: "git clone git@gitlab.com:gorails/jumpstart-pro.git YOUR_PROJECT_NAME"

Go to github and create a repo for YOUR_PROJECT_NAME

Copy the github clone info and add it as a separate remote: git remote add YOUR_PROJECT_NAME git@github.com:YOUR_GITHUB_USERNAME/YOUR_PROJECT_NAME.git

Make a change like updating the README.md and do the add / commit dance and push to your repo as git push YOUR_PROJECT_NAME master

Even though I've worked with Git for more than a decade now, I've previously only worked with Heroku as a git remote so I missed the implications of this when I originally cloned

my first JumpStart Pro app. Sigh.

The critical mistake I made was that I had started my second JumpStart Pro project by copying from the first JumpStart Pro project I had made and the NOT moving over the .git folder. This meant that my new project was now disconnected from the original remote so I couldn't just do a git pull to update it.

Here's what I did to fix this:

I started by creating a new application cloning into a new location: git clone git@gitlab.com:gorails/jumpstart-pro.git okrsnow

I created a new git repo for my app

I added the new repo with: git remote add okrsnow git@github.com:fuzzygroup/okrsnow.git

I made a change and pushed to git push okrsnow master

I manually copied over all the code from my original JumpStart pro repo. Given that Rails apps have such a clearly defined directory structure, this was tedious but not insurmountable.

I had to adapt my normal development practices so that I remember to always use the remote when working with branches.

Oh and About that Bug

I noticed that Chris fixed it within a few hours of my reporting it. Thanks Chris!

Evaluate A Reverse Polish Notation Equation With JavaScript

Nic Raboy · Nic Raboy (polyglot developer)

Previously, I demonstrated how to convert an Infix Notation expression into Reverse Polish Notation using JavaScript, but I never explained how to evaluate the expression.

Reverse Polish Notation via Wikipedia :

A mathematical notation in which every operator follows all of its operands, in contrast to Polish notation, which puts the operator in the prefix position. It is also known as postfix notation and is parenthesis-free as long as operator arities are fixed.

In this phase two article, we're going to look at how to solve a mathematical expression that has been parsed into Reverse Polish Notation (RPN).

The post Evaluate A Reverse Polish Notation Equation With JavaScript appeared first on The Polyglot Developer .

Upgrading Python and Pip and VirtualEnv on Ubuntu

Scott Johnson (fuzzyblog)

I found myself this morning trying to get a Python stack running on an Ubuntu instance only to hit this crap:

Installing collected packages: numpy, pytz, six, python-dateutil, pandas, nltk, pyparsing, cyclr, kiwisolver, matplotlib, kafka-python, pyyaml, murmurhash, zipp, importlib-metadata, catalogue, plac, tqdm, setuptools, srsly, cymem, preshed, blis, wasabi, thinc, idna, urllib3, certifi, chardet, requests, spacy, pluggy, attrs, py, more-itertools, wcwidth, packaging, pytest, langdetect, spacy-langdetect, editorconfig, jsbeautifier, docutils, jmespath, botocore, s3transfer, boto3, torch, dataclasses, tokenizers, click, joblib, regex, sacremoses, filelock, sentencepiece, transformers, scipy, scikit-learn, gunicorn, WebOb, firefly-python, pycparser, cffi, zstandard, lefthook, termcolor, fire, redis, sseclient, absl-py, protobuf, markdown, pyasn1, pyasn1-modules, cachetools, rsa, google-auth, tensorboard-plugin-wit, werkzeug, grpcio, wheel, oauthlib, requests-oauthlib, google-auth-oauthlib, tensorboard, future, pytorch-lightning

Segmentation fault (core dumped)

So I guess it is time to do the Python upgrade dance. Here you go:

```
pip3 install --upgrade pip
sudo apt-get update && sudo apt-get upgrade
sudo apt-get install python3.7
sudo pip3 install --upgrade virtualenv
```

Pip upgrade .

Use RegEx To Test Password Strength In JavaScript

Nic Raboy · Nic Raboy (polyglot developer)

Recently one of my Twitter followers asked me how they might validate password strength using regular expressions (RegEx) in their code.

Regular expressions via Wikipedia :

A sequence of characters that forms a search pattern, mainly for use in pattern matching with strings, or string matching.

RegEx is nice because you can accomplish a whole lot with very little. In this case we are going to check various aspects of a string and see if it meets our requirements, being a strong or medium strength password.

The post Use RegEx To Test Password Strength In JavaScript appeared first on The Polyglot Developer .

Prerender Your AngularJS Apps To Boost SEO

Nic Raboy · Nic Raboy (polyglot developer)

A Twitter subscriber of mine recently asked me if I knew how to use Prerender with my AngularJS applications to boost SEO when being crawled by search engines. My immediate answer was no, but I offered to help him figure it out and write an article on it.

Prerendering AngularJS web apps is critical for SEO because when web crawlers such as Google crawl a web page, they do not process all the AngularJS source code and curly bracket elements. With prerendering enabled, a web crawler will see the actual value found in curly brackets rather than the variables themselves.

The post Prerender Your AngularJS Apps To Boost SEO appeared first on The Polyglot Developer .

Building Amazon Alexa Skills With Node.js, Revisited

Nic Raboy · Nic Raboy (polyglot developer)

A little more than two years ago, when the Amazon Echo first started picking up steam and when I was first exposed to virtual assistants, I had written a tutorial around creating a Skill for Amazon Alexa using Node.js and simple JavaScript. In this tutorial titled, Create an Amazon Alexa Skill Using Node.js and AWS Lambda , we saw how to create intent functions and sample utterances in preparation for deployment on AWS Lambda. I later wrote a tutorial titled, Test Amazon Alexa Skills Offline with Mocha and Chai for Node.js , which focused on building unit tests for these Skills and their intent functions. Fast forward to now and a few things have changed in the realm of Skill development.

In this tutorial we're going to see how to build a Skill for Alexa powered devices using Node.js and test it using popular frameworks and libraries such as Mocha and Chai .

The post Building Amazon Alexa Skills With Node.js, Revisited appeared first on The Polyglot Developer .

ruby invalid option dash colon Or When Ruby Goes Insane in the Brain

Scott Johnson (fuzzyblog)

This is a short one but a weird one. My main box rebooted today and when I went to create a blog post, I got this madness:

```
> jekyll post "A Social Media Marketing Strategy that Mom Would Approve"
ruby: invalid option -: (-h will show valid options) (RuntimeError)
```

In all my years of ruby, I've never seen -: . All the normal jiggery pokery of:

which ruby

rvm list

etc failed to give me any insights. I finally found the answer on an obscure Github issue that said "unset RUBYOPT" so:

```
> unset RUBYOPT
```

And that gave me back:

```
blog on j gh-pages [x] via v2.3.1 on us-west-2 > ruby
-version ruby 2.3.1p112 (2016-04-26 revision 54768) [x86_64-darwin16]
```

```
blog on j gh-pages [x] via v2.3.1 on us-west-2 > jekyll
post "A Social Marketing Strategy that Mom Would Approve"
New post created at _posts/2020-03-24-a-social-marketing-
strategy-that-mom-would-approve.md.
```

And now, once again, I can (joyfully) write!

Restart ID sequences in Postgres on Truncation

Scott Johnson (fuzzyblog)

Note : This is from a real world thing that I'm about to release.

Like a lot of database apps in the world, mine generally tend to use auto incrementing ids for object identifiers. An object identifier is a value in a system which uniquely identifies something. Let's say you have a url like this:

```
http://localhost:3169/
locations?country_id=91
```

What that is saying, under the hood, is "Get the data from the locations store and give me the object with id value 91". The thing that I'm currently building is populated with a bunch of data which is coming from CSV files that I'm loading from the Internet and I need to wipe the data in the system every time it rebuilds. Now when I say location store, I mean, in this case, a locations table in a Postgres database. When you wipe the contents of a table in Postgres you end up with the next object in the table getting the last ID value plus 1.

The secret is to add this snippet to your truncation routine:

```
RESTART IDENTITY
```

or in full:

```
TRUNCATE states RESTART
IDENTITY
```

And in Rails I have this implemented on all my core classes as:

```
def self . truncate ActiveRecord :: Base . connection . execute ( "TRUNCATE
```

```
states RESTART IDENTITY" ) end
```

Since this is a def self, think of it as a C style static thing. By defining it this way, I can simply call it as State.truncate to eliminate all data in my states store (ok its is a table but it could be more complex) and restart my ID sequences. And in cases it wasn't clear exactly why you want this, since I'm processing the same data files in Production as in Development, my bookmarks become consistent allowing easier debugging.

Create a Custom Stream Elements Overlay for YouTube Subscribers

Nic Raboy · Nic Raboy (polyglot developer)

I recently got into streaming video games on a consistent basis and needed a better call to action (CTA) to encourage subscribers. In an attempt to encourage more subscriptions, I thought it'd be beneficial to show to my viewers how many current subscriptions I had and have them watch the number increase if they choose to subscribe.

While I could just show static plain text of this, I thought it'd be more professional to have it nicely formatted and updating live. For this I stumbled upon Stream Elements .

With Stream Elements I could create web overlays to show on my stream. However, I was responsible with the design of these overlays.

In this tutorial, I'm going to share how I created a custom widget for showing total YouTube subscribers on my stream.

The post Create a Custom Stream Elements Overlay for YouTube Subscribers appeared first on The Polyglot Developer .

So That One Time You Played With Docker

Scott Johnson (fuzzyblog)

layout: post title: So That One Time You Played with Docker for hte

So that one time, at band camp – wait; wrong movie, this is a technical blog post about Docker ...

I recently helped a friend doing some federal work in the medical field get a better understanding of Docker and how Docker can mess about with open source licensing. We got up a “open source” stack of tooling using a bunch of heavy weight tools:

A big Python app

Rabbit MQ

And then, as you so often do, we finished up and I got distracted by the next bit of crazy nerd-fu that runs about in my life (server down; wifi crisis; son needing help with his PC; who can remember) and then I went back to my desk, closed my terminal prompt and never thought much more about this.

Lately I've been noticing some slowness on my machine and I just happened to run docker ps and what do I see but:

```
docker ps CONTAINER
ID IMAGE COMMAND
CREATED STATUS PORTS
NAMES b93fbdc58d2
data_streamer_to_sqs_ruby
“ruby ./main.rb” 29 sec-
onds ago Up 27 seconds
nice_torvalds 605da7fe5159
dsarchive/dsa_girder:latest
“/bin/sh -c ‘sudo -
E...” 2 weeks ago Up
3 hours 0.0.0.0:8080-
>8080/tcp dsa_girder
0b0eb7730fb0 dsarchive/
dsa_worker:latest “/bin/
```

```
sh -c ‘sudo -E...” 2
weeks ago Up 3 hours
dsa_worker df6fd902c11b
memcached:latest “docker-
entrypoint.s...” 2 weeks
ago Up 3 hours
11211/tcp dsa_memcached
dc7f57778b4e mongo:latest
“docker-entrypoint.s...”
2 weeks ago Up
3 hours 27017/tcp
dsa_mongodb 863768f4f33f
rabbitmq:management
“docker-entrypoint.s...” 2
weeks ago Up 3 hours
4369/tcp, 5671-5672/tcp,
15671-15672/tcp, 25672/tcp
dsa_rabbitmq
```

Yep. That's the whole set of processes that I was using for my federal friend. Sigh. They've been happily sucking up CPU for more than 2 weeks. grumble

The solution:

```
docker stop 863768f4f33f
dc7f57778b4e df6fd902c11b
0b0eb7730fb0 605da7fe5159
```

Making Ruby Scripts work Under System D Using rbenv

Scott Johnson (fuzzyblog)

This has bit me I think every single time I make a SystemD launch script using ruby and rbenv (RVM is worse; imho). Perhaps writing it down for once makes the lesson stick.

Let's start with our SystemD unit file that doesn't work:

```
[Unit] Description=Twitter To Kafka
```

```
[Service] SyslogIdentifier=twitter_to_kafka User=ubuntu PIDFile=/home/ubuntu/twitter_to_kafka/twitter_to_kafka.pid WorkingDirectory=/home/ubuntu/twitter_to_kafka
```

```
ExecStart=/home/ubuntu/twitter_to_kafka/twitter_to_kafka.sh ExecReload=/home/twitter_to_kafka/twitter_to_kafka/twitter_to_kafka.sh ExecStop=/bin/kill -s QUIT $MAINPID Restart=always
```

```
[Install] WantedBy=multi-user.target
```

This generates errors like these:

```
> sudo systemctl status twitter_to_kafka.service
● twitter_to_kafka.service
- Twitter To Kafka
Loaded: loaded (/lib/systemd/system/twitter_to_kafka.service; enabled; vendor preset: enabled)
Active: failed (Result: exit-code) since Thu 2020-02-13 12:57:48 UTC; 1min 54s ago
Process: 7615 ExecStart=/home/ubuntu/twitter_to_kafka/twitter_to_kafka.sh (code=exited, status=203/EXEC)
Main PID: 7615
```

```
(code=exited, status=203/EXEC)
```

```
Feb 13 12:57:48 ip-172-31-24-213 systemd[1]: twitter_to_kafka.service: Main process exited, code=exited, status=203/EXEC
```

```
Feb 13 12:57:48 ip-172-31-24-213 systemd[1]: twitter_to_kafka.service: Failed with result 'exit-code'.
```

```
Feb 13 12:57:48 ip-172-31-24-213 systemd[1]: twitter_to_kafka.service: Service hold-off time over, scheduling restart.
```

```
Feb 13 12:57:48 ip-172-31-24-213 systemd[1]: twitter_to_kafka.service: Scheduled restart job, restart counter is at 5.
```

```
Feb 13 12:57:48 ip-172-31-24-213 systemd[1]: Stopped Twitter To Kafka.
```

```
Feb 13 12:57:48 ip-172-31-24-213 systemd[1]: twitter_to_kafka.service: Start request repeated too quickly.
```

```
Feb 13 12:57:48 ip-172-31-24-213 systemd[1]: twitter_to_kafka.service: Failed with result 'exit-code'.
```

```
Feb 13 12:57:48 ip-172-31-24-213 systemd[1]: Failed to start Twitter To Kafka.
```

The problem is that rbenv is a shell extension so the shell has to be loaded to run this correctly. Just adding /

Check Network Connection With Ionic Framework

Nic Raboy · Nic Raboy (polyglot developer)

When creating a mobile app, specifically one that makes heavy use of the internet, it is often necessary to make sure an internet connection exists at launch and possibly display a message or perform an action if one does not exist.

The following is for an Ionic Framework 1 application. If you're using Ionic 2, you will want to check here .

The following chunk of example code is typically found in your Ionic Framework project's js/app.js file. It demonstrates how to check if a network connection exists on application load and how to display an Ionic popup that is nicely styled in comparison to a JavaScript alert dialog.

The post Check Network Connection With Ionic Framework appeared first on The Polyglot Developer .

Track The Device Geolocation In A Native-Script Angular Mobile Application

Nic Raboy · Nic Raboy (polyglot developer)

I get a particular set of questions quite a bit on my blog and other social media outlets. One of these questions includes how to use geolocation features such as GPS tracking within a NativeScript mobile application for iOS and Android. Many people want to be able to gather location information and in many cases use this information for mapping.

So what does it take to make use of the device GPS hardware for location tracking?

We're going to see how to create a mobile application for Android and iOS using NativeScript and Angular that makes use of geolocation in a few different ways.

The post [Track The Device Geolocation In A NativeScript Angular Mobile Application](#) appeared first on [The Polyglot Developer](#).

Query A GraphQL API With Vue.js And Axios

Nic Raboy · Nic Raboy (polyglot developer)

I've written quite a bit of content around developing a GraphQL API with various programming technologies such as Node.js, Golang, and Java. Heck, I even wrote an eBook on the subject titled, [Web Services for the JavaScript Developer](#). However, I haven't really produced any content around interacting with those APIs using modern frameworks and client facing technologies, only cURL and Postman.

So how do you interact with a GraphQL API using something like Angular, React, or Vue.js?

There are a lot of frameworks to cover, but in this particular tutorial we're going to see how to use Vue.js and simple JavaScript to interact with a GraphQL API.

The post [Query A GraphQL API With Vue.js And Axios](#) appeared first on [The Polyglot Developer](#).

Using PassportJS In Your Node.js And Couchbase Web App

Nic Raboy · Nic Raboy (polyglot developer)

Why I've Gone Hybrid Over Native For Mobile Apps

Nic Raboy · Nic Raboy (polyglot developer)

I started developing mobile Android applications in late 2011. It wasn't until 2012 that I released my first application to the Android Marketplace, now known as the Google Play Store. Up until early 2014, I only made native Android applications using the official Android SDK. As of now, I have only been using web technologies to create mobile hybrid applications for Android and iOS.

The following will explain why I've chosen to develop using modern web technologies rather than native mobile languages.

The post [Why I've Gone Hybrid Over Native For Mobile Apps](#) appeared first on [The Polyglot Developer](#).

TPDP Episode #11: Continuous Integration And Deployment For The Polyglot Developer

Nic Raboy · Nic Raboy (polyglot developer)

Continuous integration (CI) and continuous deployment (CD) are terms that I hear thrown around quite frequently. I've been a software developer for a long time, but it is only recently that I've welcomed these terms into my life. CI and CD is the automated process of running various tasks such as unit testing or building a version controlled project. In this episode of [The Polyglot Developer Podcast](#), I have guest speaker Ivan Nemytchenko, from [GitLab](#), helping me explain what continuous integration is all about and why [GitLab](#) has gone ahead and made a completely free set of tools around it.

The post [TPDP Episode #11: Continuous Integration And Deployment For The Polyglot Developer](#) appeared first on [The Polyglot Developer](#).

Create Circular Images With Curved Text In Affinity Designer

Nic Raboy · Nic Raboy (polyglot developer)

I was at an event not too long ago where someone recommended I take a look at PassportJS for my Node.js applications. In case you're unfamiliar with PassportJS, it is an authentication middleware that makes it easy to work with logins, whether they are with basic login, or with social media accounts such as Facebook and Twitter.

Being that I'm into different authentication techniques, as seen in my other OAuth articles, I figured it wouldn't be a bad idea to do a walkthrough on PassportJS with Express Framework for Node.js.

The post Using PassportJS In Your Node.js And Couchbase Web App appeared first on The Polyglot Developer .

Animate HTML Components In An Angular Application

Nic Raboy · Nic Raboy (polyglot developer)

When building a web application, not only do you need amazing features, but you also need a stunning user experience and that can often be managed by making good use of UI components. When I'm using my favorite web applications, the way components are displayed on the screen are sometimes what makes the application so memorable. Designers will often animate components to make them stand out.

When building a web application you can use CSS keyframe animations, but if you're using Angular, there is another way. We're going to see how to use the Angular Animations component to animate UI components on the screen.

The post Animate HTML Components In An Angular Application appeared first on The Polyglot Developer .

Create An Email Subscription Popup With jQuery

Nic Raboy · Nic Raboy (polyglot developer)

As you'll recall, The Polyglot Developer was once using WordPress. Back when I was using WordPress, I was using a plugin called Icegram, which is similar to SumoMe and OptinMonster, but it allowed me to present popups after a period of time to prompt users to subscribe to my newsletter.

I get that not everyone appreciates an annoying popup, but it was great for me because I was getting a lot of new email subscribers very quickly.

Since having dropped WordPress, I haven't been able to find a plugin that offered similar functionality to what I had. This inspired me to create my own plugin using jQuery that I could use anywhere, including my Hugo powered blog. We're going to see how to create our own opt-in style popup for collecting email leads.

The post Create An Email Subscription Popup With jQuery appeared first on The Polyglot Developer .

As many of you know, I've been trying to expand the brand for The Polyglot Developer. I've redesigned the site, printed shirts, and even started creating stickers.

I've been printing circular stickers with Sticker Mule, and one thing that had been a real challenge is adding circular or curved text to them. It is nice to have the brand logo on the stickers, but until developers know what the logo is, it helps to have a website link as well.

This led me to Affinity Designer, the much cheaper alternative to Adobe Illustrator, and the more polished alternative to Inkscape. We're going to see how to create some nice images with curved text in Affinity Designer.

The post Create Circular Images With Curved Text In Affinity Designer appeared first on The Polyglot Developer .

Syncing Data With Dropbox Using Ionic Framework

Nic Raboy · Nic Raboy (polyglot developer)

Storing or syncing data remotely is often a need in modern apps. This generation is all about the cloud and how to be a part of it. Ionic Framework is a great way to make hybrid cross platform mobile Android and iOS applications, but how about sharing data between them? Dropbox is one of a few cloud storage services that happens to have great APIs and documentation.

The following won't necessarily show you the best way for syncing data in Ionic Framework using Dropbox's Datastore API, but it will show you how I did it with success.

The post Syncing Data With Dropbox Using Ionic Framework appeared first on The Polyglot Developer .

I write a lot of code and I'll confess something – my code is messy.

Scott Johnson (fuzzyblog)

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